



Examiners' Report

Principal Examiner Feedback

June 2025

Pearson BTEC Nationals

In Applied Science/Forensic and Criminal Investigation
Unit 1: Principles and Applications of Science I (31617H)

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Grade Boundaries

What is a grade boundary?

A grade boundary is where we set the level of achievement required to obtain a certain grade for the externally assessed unit. We set grade boundaries for each grade, at Distinction, Merit and Pass.

Setting grade boundaries

When we set grade boundaries, we look at the performance of every learner who took the external assessment. When we can see the full picture of performance, our experts are then able to decide where best to place the grade boundaries – this means that they decide what the lowest possible mark is for a particular grade.

When our experts set the grade boundaries, they make sure that learners receive grades which reflect their ability. Awarding grade boundaries is conducted to ensure learners achieve the grade they deserve to achieve, irrespective of variation in the external assessment.

Variations in external assessments

Each external assessment we set asks different questions and may assess different parts of the unit content outlined in the specification. It would be unfair to learners if we set the same grade boundaries for each assessment, because then it would not take accessibility into account.

Grade boundaries for this, and all other papers, are on the [Pearson grade boundaries page](#).

Introduction

Biology

This paper is one of three externally assessed papers worth 30 mark each that assess Unit 1 – Principles and Applications of Science I. This paper covers key biological concepts including the structure and functions of cells and tissues, including specialised cells. The paper also covered atherosclerosis, the specialisation of white blood cells and muscles. Learners showed that they understand the relationship between the structure and function of cells and tissues throughout the unit.

Chemistry

This paper is one of three externally assessed papers, worth 30 marks each, that assess Unit 1 – Principles and Applications of Science 1. This paper covers key chemistry concepts including atomic structure and bonding and chemical and physical properties of substances and their uses.

Physics

The examination was designed to allow learners to show their knowledge and understanding of 'Waves in Communication'. The examination was comparable to those set previously as there were no changes to the specification or the composition of the paper. The topics included fundamentals of waves and wave behaviour, interpretation of graphical information, calculations using the wave speed and the inverse square law equations. Applications included endoscopes and stringed musical instruments.

Overall Performance of the Unit

Biology

Learners still struggle to recall definitions and showed a lack of understanding of cell theory. Rather than a simple recall of the definition of tissue and organ, which learners had been asked previously, the topic of cell theory was the focus of the extended open response question, so learners had a different opportunity to demonstrate their knowledge of key word definitions. Unfortunately, the majority of learners did not seem to use the

question prompt and did not explain cell theory, instead they focused on the images in the figure and attempted to just describe skin and blood. However, the majority of learners were able to access the extended open response question and gain at least one mark in level 1 for their knowledge of the function of blood cells. Learners were able to perform the calculation given using clear and logical working. The additional guidance document can be found on the Applied Science website and should help teachers understand the depth to which they need to teach the content for the exams for unit 1 in conjunction with the specification and pass question papers and mark schemes.

Chemistry

Learners that were successful this year showed a good understanding of key terms, scientific vocabulary and mathematical skills and were able to apply this knowledge and understanding to new scenarios. They engaged with the question and used the prompts in the questions to inform their answers. They were precise with their language, were able to link ideas and explain in detail where necessary. Learners that performed well, were able to write balanced equations, explain differences in strength of bonds of compounds and differences in melting points of different types of elements. They were able to carry out calculations in a clear and logical manner, showing their working at all points, which helped them to arrive at the correct answer.

Physics

Learners were generally able to identify that water waves are transverse waves, digital signals consist of an encoded series of zeroes and ones, and most could also identify the amplitude and time period of a wave from a graphical representation.

Learners were able to demonstrate their ability to interpret a graph showing how percentage transmission through the atmosphere and wavelength of microwaves are related, and could accurately describe the trend shown on the graph. Many learners were able to correctly complete calculations, although some learners still had difficulty in correctly rearranging equations, with some also struggling in being able to correctly use standard form and perform more complex mathematical operations. Most learners could correctly state that a ray of light will refract as it travels from air to glass, and could describe what happens during refraction. Knowledge of how

endoscopes work was not well understood, and many learners struggled to explain why low frequency radio waves do not travel out of the atmosphere.

Although learners were familiar with the idea that detection of a signal is stronger the closer a detector is to the transmitter, many struggled to explain why this is so.

Many learners were able to discuss how pulling a bow across the strings of a violin makes a sound, but few were able to relate this to the topic of stationary waves and harmonics, with fewer still able to correctly use technical terminology to explain how a stationary wave is established in these conditions.

Individual Questions

Biology

Question 1

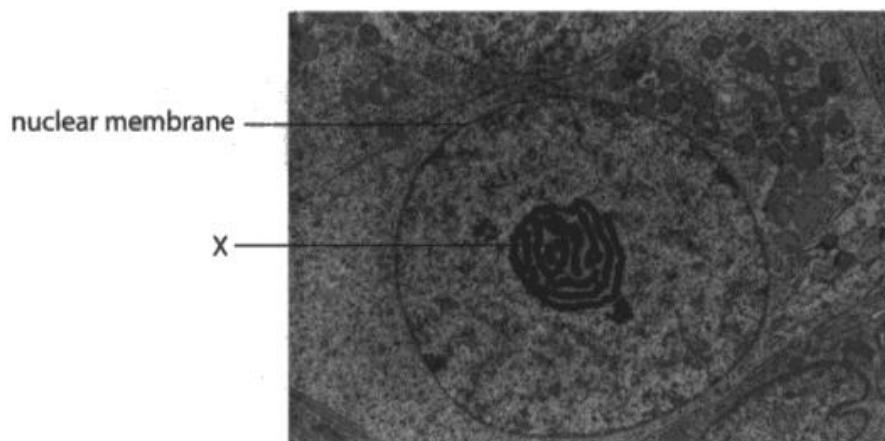
1(a)(i)

The expected response was “nucleolus.” Although “nucleus” appeared frequently, this could not be credited as the figure clearly indicated a distinct region within the nucleus. Answers such as “DNA” or “chromosome” were accepted. A number of learners appeared unfamiliar with the structure, offering unrelated organelle names including SER, RER, or Golgi apparatus. The term “nucleoid” also appeared more than once, likely due to confusion with prokaryotic cell structures.

The following response was awarded 1 mark.

1 Animal cells and plant cells are eukaryotic cells.

Figure 1 shows an electron micrograph of a cross section of a nucleus.



(Source: © PAL)

Figure 1

(a) (i) Name the structure labelled X in Figure 1.

(1)

nucleolus

1(a)(ii)

This question asked learners to identify whether chloroplasts, plasmodesmata, and ribosomes are found in animal cells, plant cells, or both. Responses for chloroplasts and ribosomes were generally accurate, indicating sound understanding of these organelles. While many learners achieved two or three marks, suggesting a reasonable grasp overall, errors were most commonly associated with plasmodesmata. These included incorrectly selecting both cell types or identifying plasmodesmata as present in animal cells only, reflecting some confusion around plant-specific cellular structures.

The following response was awarded 3 marks.

The learner had correctly ticked 'found in a plant cell' for the chloroplasts. The incorrect tick for the plasmodesmata had been crossed out and so the mark could be awarded. The learner had corrected to both 'found in animal cell' and 'found in plant cell' for the ribosomes.

- (ii) Complete Table 1, using ticks to show which organelles are found in animal cells and which organelles are found in plant cells.

(3)

The first row has been completed.

organelles	found in an animal cell	found in a plant cell
Golgi apparatus	✓	✓
chloroplasts		✓
plasmodesmata	✗	✓
ribosomes	✓	✓

Table 1

The following response was awarded 1 mark.

For chloroplasts the learner had ticked 'found in a plant cell', which is correct. For plasmodesmata they have ticked 'found in an animal cell', which is incorrect. For ribosomes they have ticked 'found in an animal cell', but they should tick both, therefore they do not get this mark.

- (ii) Complete Table 1, using ticks to show which organelles are found in animal cells and which organelles are found in plant cells.

(3)

The first row has been completed.

organelles	found in an animal cell	found in a plant cell
Golgi apparatus	✓	✓
chloroplasts		✓
plasmodesmata	✓	
ribosomes	✓	

Table 1

1(b)

This was a multiple-choice item assessing learners' ability to recall the equation for calculating magnification. While this concept has previously been tested through numerical calculations, the format of this question

enabled learners who may find mathematical manipulation challenging to demonstrate understanding of the relationship between image size, actual size, and magnification.

The following response was awarded 1 mark.

Microscopes magnify the size of actual objects so that they can be viewed as an image in greater detail.

(b) Which equation is used to calculate magnification?

(1)

- ☒ A $\frac{\text{size of actual object}}{\text{size of image}}$
- ☒ B size of actual object $\times$ size of image
- ☒ C $\frac{\text{size of image}}{\text{size of actual object}}$
- ☒ D size of image + size of actual object



Question 2

2(a)

This question required the learners to interpret an ECG trace and identify which section of the ECG trace indicated the contraction of the ventricles. The majority of learners correctly identified section B.

The following response was awarded 1 mark.

2 Doctors use electrocardiograms (ECGs) to diagnose cardiovascular diseases.

Figure 2 shows an ECG trace.

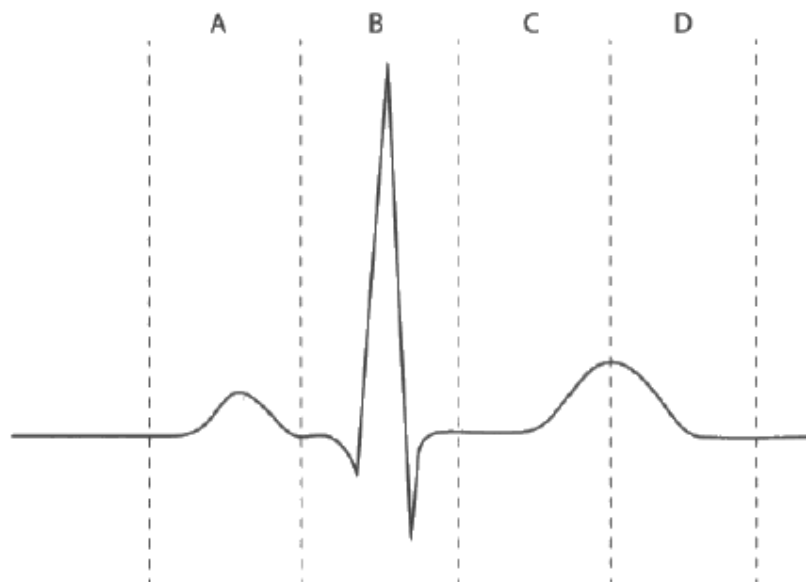


Figure 2

(a) Identify the part of the ECG trace that indicates the contraction of the ventricles.

(1)

- ☒ **A** part A
- ☒ **B** part B
- ☒ **C** part C
- ☒ **D** part D

2(b)(i)

This question focused on dietary factors that could lead to atherosclerosis. The question was generally well answered overall, with many learners correctly identifying fats in various forms. Sugars also featured prominently in responses. A number of learners referred instead to lifestyle choices such as smoking, alcohol consumption, and exercise, suggesting some misinterpretation of the question. Where food types were named, common examples included fast food and junk food, which were not specific enough about the component of the diet and so could not be credited.

The following response was awarded 0 marks.

This is a common error. The learner has given a lifestyle factor and not a component of the diet and so the mark cannot be awarded. We ignored all lifestyle factors as they are not contradictions of the correct answers and so would not negate the mark if given in a list.

Atherosclerosis is the build-up of plaque in the arteries and can be caused by an unhealthy diet.

(b) (i) State **one** component in a person's diet that can cause plaque to form.

(1)

Smoking

The following response was awarded 1 mark. Both answers would have been acceptable.

Atherosclerosis is the build-up of plaque in the arteries and can be caused by an unhealthy diet.

(b) (i) State **one** component in a person's diet that can cause plaque to form.

(1)

too many fats / cholesterol

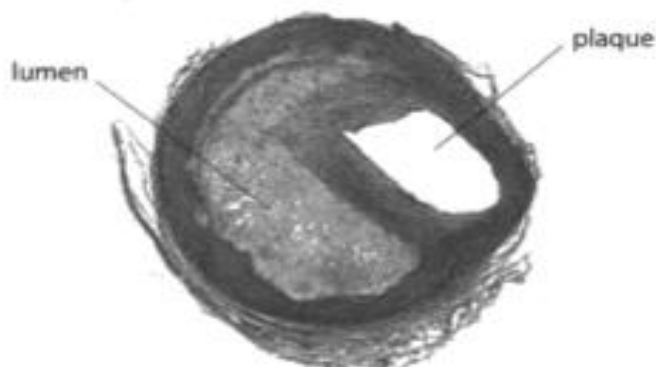
2(b)(ii)

Learners approached the percentage decrease calculation using a range of methods, with most opting for method 1. A number successfully calculated the correct value, though rounding errors were common, particularly rounding 30.95 to 30.9 or 30. The most frequent error involved using an incorrect denominator, such as dividing by 2.9 instead of 4.2. Learners who showed working were able to access marking point 1, although a significant minority did not present any working. Multiple methods were often seen within individual responses. An increased number attempted method 2, but few completed it fully; more often, they achieved only the first marking point. A small number appeared not to have access to a calculator and relied on informal chunking strategies, which typically led to imprecise outcomes.

The following response was awarded 1 mark.

Without the calculation of a difference, we presumed that the learner was using the second method in the mark scheme. The learner has completed the calculation for MP1, dividing the new by the original value and multiplying by 100. The learner should have then completed the second marking point (100- their value) and then given the evaluation of 31 to gain the 3 marks.

Figure 3 shows the lumen of an artery with a plaque.



(Source: © PAL)

Figure 3

The lumen of the artery originally had a diameter of 4.2 mm.

The artery developed a plaque.

The lumen of the artery then had a diameter of 2.9 mm.

(ii) Calculate the percentage decrease in the diameter of the lumen.

(3)

Show your working.

$$\frac{2.9}{4.2} = 0.69$$

$$0.69 \times 100 = 69$$

$$\frac{\text{New}}{\text{Old}} \times 100$$

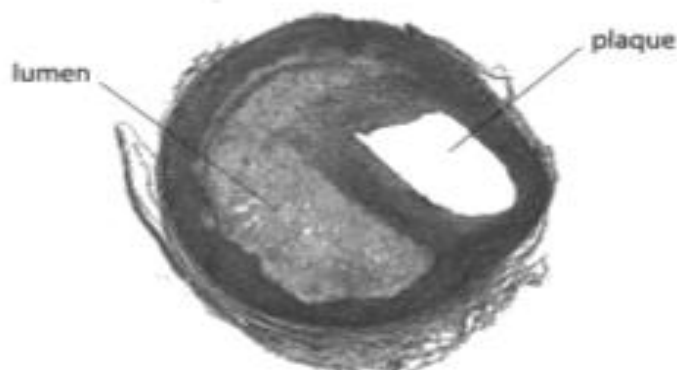
percentage decrease in the diameter of the lumen = 69 %

The following response was awarded 3 marks.

The learner has given the correct response, correctly rounded from the calculated decimal, and so the 3 marks were awarded. The learner showed their working out and therefore any process marks could be awarded, if their final answer was not correct. It is good practice for learners to write down

their working out to allow for error carried forward (ECF) marks in calculations.

Figure 3 shows the lumen of an artery with a plaque.



(Source: © PAL)

Figure 3

The lumen of the artery originally had a diameter of 4.2 mm.

The artery developed a plaque.

The lumen of the artery then had a diameter of 2.9 mm.

(ii) Calculate the percentage decrease in the diameter of the lumen.

(3)

Show your working.

$$\frac{2.9}{4.2} \times 100 = 69.047619$$

$$100 - 69.047619 = 30.95238095$$

percentage decrease in the diameter of the lumen = 30.95 %

2(b)(iii)

Responses to this question indicated a widespread lack of understanding. Many learners incorrectly identified plaque as a pathogen and described the role of white blood cells (WBCs) in phagocytosis or antibody production. Others misinterpreted plaque as the result of repeated phagocytosis and WBC death or confused it with dental plaque. While a few learners demonstrated a basic grasp of atherosclerosis, most failed to access key marking points, particularly those involving WBC movement to damaged tissue. Marking points 2 and 3 were the most commonly achieved, though imprecise terminology (e.g. "eating cholesterol" or "producing foam cells") often hindered credit. Misconceptions around foam cells were evident, with WBCs described as producing rather than becoming foam cells. Inflammatory processes were occasionally mentioned, but references to apoptosis or the cascade effect were notably absent.

The following response was awarded 1 mark.

The reference to white blood cells getting to the site of the plaque formation is sufficient for MP1.

(iii) Explain how white blood cells are involved in the formation of plaques.

(3)

↳ white blood cells go to site that
damaged
↳ too many white blood cells so it
starts building up
↳ build up eventually turns into
plaque

The following response was awarded 2 marks.

The learner has not clearly stated that the white blood cells have moved to the site of damage although their answer does imply this as it is written about the action at the plaque. We want the idea of moving to, rather than just being there for the award of MP1. The learner has then given MP3 for stating absorbing cholesterol and then MP2 for the white blood cells becoming foam cells.

(iii) Explain how white blood cells are involved in the formation of plaques.

(3)

When there is inflammation on endothelium lining of arteries. White blood cells such as macrophages will absorb ~~and~~ LDL cholesterol in the blood at the point of inflammation. The macrophages will then form foam cells which will be built up on the endothelial layer. This then causes minerals like calcium to also build up to form plaque. (Total for Question 2 = 8 marks)

Question 3

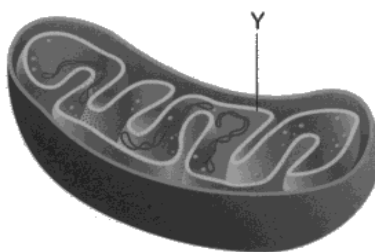
3(a)

The correct answer was "cristae". While many learners identified that the structure related to a membrane, the majority gave non-specific responses such as "membrane," or "folded membrane", which was not sufficient for the mark. Only a small number recognised it as the "inner membrane." "Mitochondria" was a frequent response, suggesting that the learners had not read the stem of the question, as the mitochondria had been named within the stem of the question. Other incorrect answers included unrelated cellular components, with "matrix" appearing notably often. These patterns suggest a general awareness of mitochondrial structure, but limited familiarity with the specific terminology required.

The following response was awarded 1 mark.

3 Muscle cells have more mitochondria than most other cell types.

Figure 4 shows a diagram of a mitochondrion.



(Source: © PAL)

Figure 4

(a) Name structure Y in Figure 4.

(1)

~~matrix~~ cristae

3(b)

This question revealed limited understanding of the role of ATP in muscle contraction. A significant number of learners scored no marks, often failing to use key terminology such as actin, myosin, or power stroke. Many responses echoed the question stem by stating that ATP releases energy to cause movement, few could articulate how this energy is utilised in the contraction process. Marks were most frequently awarded for identifying myosin attachment to actin (MP1). Some learners achieved MP2, with references to both lateral head swing and actin glide seen. A small number of learners accessed MP3, by referencing the detachment of myosin.

The following response was awarded 0 marks.

This was a common mistake – the learner has repeated the stem and question rather than describing how ATP is used in myofibril contraction.

The mitochondria in muscle fibres produce ATP.

ATP is used as the energy source for myofibril contraction.

(b) Describe the role of ATP in myofibril contraction.

(2)

~~ATP is produced~~ ATP provides energy in myofibril contractions. Allowing the muscles to have enough energy to work, which is why have lots of mitochondria, as provides lots of ATP, giving energy.

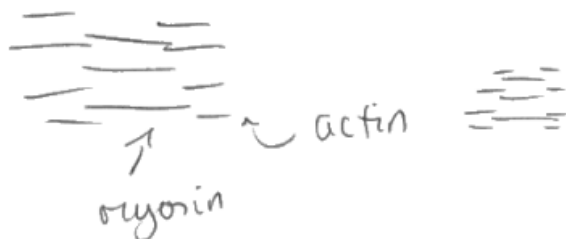
The following response was awarded 2 marks.

Diagram not needed on this occasion as the learner had expressed the marking points in their written response, but marks could be given for diagrams.

(b) Describe the role of ATP in myofibril contraction.

(2)

~~ATP~~ There is an action potential where myosin and actin form a crossbridge and the ^{myosin} ~~actin~~ moves and slides the ^{actin} ~~myosin~~. ATP breaks this bond between myosin and actin, breaking the crossbridge. Allowing for the contraction to be repeated.



3(c)

The concept of a "dense network of capillaries" was not well understood by many learners, resulting in a high number of zero mark responses. While most recognised that blood delivers oxygen for respiration, marks were frequently lost due to lack of precision, for instance, omitting references to a high concentration of oxygen or failing to specify aerobic respiration. Vague statements such as avoiding fatigue or enabling marathon running were common.

The following response was awarded 1 mark.

The learner has not given sufficient detail about the oxygen or glucose in the blood or referring to aerobic respiration.

People with a higher percentage of slow-twitch muscle fibres are better at running marathons than sprinting short distances.

(c) Explain why slow-twitch muscle fibres need a dense network of capillaries.

(3)

Slow twitch need many capillaries because it's used over a period of time (long term exercise) which requires a lot of energy → from the mitochondria which requires a lot of blood flow.

(Total for Question 3 = 6 marks)

The following response was awarded 2 marks.

People with a higher percentage of slow-twitch muscle fibres are better at running marathons than sprinting short distances.

(c) Explain why slow-twitch muscle fibres need a dense network of capillaries.

(3)

slow twitch muscles need a dense network of capillaries because they produce energy through aerobic respiration, meaning they need lots of oxygen from the blood in the capillaries

Question 4

4(a)

This item was not well answered overall. The most frequent responses either repeated the concept of phagolysosome formation or stated that the pathogen was digested and so had not considered the stage following the gap. Many learners used vague or inaccurate terminology, referring to pathogens being “killed,” “broken down,” or “destroyed,” rather than correctly identifying engulfment as the key process. While “engulfed” was the most common correct term seen, the prevalence of imprecise language indicates a need to reinforce accurate scientific terminology. A number of responses also mistakenly focused on antibody production, suggesting some confusion between innate and adaptive immune processes.

The following response was awarded 1 mark.

4 White blood cells protect the body from pathogens.

Phagocytes are one type of white blood cell.

Phagocytes can digest pathogens through a process called phagocytosis.

Figure 5 shows some of the steps involved in phagocytosis.

(a) Complete Figure 5.

(1)

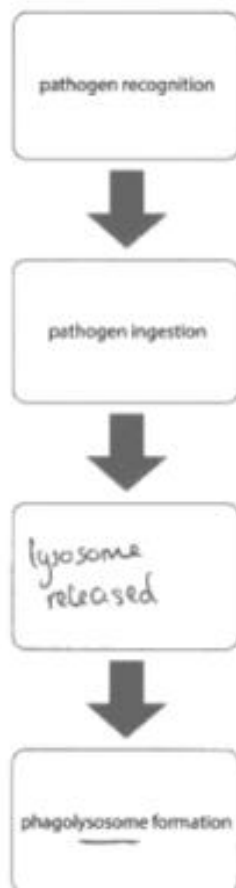


Figure 5

4(b)

Learners generally found this question challenging, with many struggling to gain credit despite having some understanding of memory cells and antibodies. A significant number of responses focused on describing the graphical trend rather than offering explanations, indicating a lack of alignment with the command verb. Most answers centred on antibody quantity rather than the speed of production, resulting in frequent omission of MP4, which required reference to the faster secondary response. MP1 was also commonly missed, as learners tended to repeat information from the stem rather than explaining the low antibody levels during the primary exposure. While many learners were able to recall that memory cells remain in the blood following infection, few gained MP2 or MP3 due to a lack of detail on how memory cells are produced or their differentiation into plasma cells. Overall, responses were often descriptive rather than explanatory, with

limited understanding shown of B cell maturation pathways and their role in adaptive immunity.

The following response was awarded 1 mark.

The learner has given MP1 as they have said the antibody concentration is low as the pathogen has never encountered this pathogen. There is no reference to the rapid production of antibodies during the second exposure for MP4. This question is not about describing the trend in the graph; an explanation to fulfil the demand of the command verb was required.

(b) Explain the differences in antibody concentration after first and second exposure.

(4)

Use information in Figure 6 to support your answer.

• After first exposure the antibody concentration was low since the primary immune response was low
• this would've been due to the fact that the lymphocytes/antibodies have never encountered this pathogen, therefore have no immediate reaction/response

• However, after second exposure the antibody concentration was high and the secondary immune response was high.

(Total for Question 4 = 5 marks)

^{qph} This would've been due to the fact that the body was now familiar to the pathogen, therefore had a prepared response to fight it off

The following response was awarded 3 marks.

There is no explanation for the low concentration during the primary immune response and so MP1 cannot be awarded. However the learner has then given sufficient detail about B-cells differentiating into memory cells for MP2 and detail about the memory cells rapidly differentiating into plasma cells for MP3. The learner has then gained MP4 for reference to the rapid production of antibodies during the secondary immune response.

In the first exposure there are a very few amount of white antibodies which due to low numbers are not effective.

The primary immune response is where the phagocyte has ingested and digested the pathogen turning into an Antigen presenting cell which produces T Helper cells which stimulate Cytotoxic T cells, which then release B plasma cells and B memory cells. The B plasma cells can then produce antibodies that can bind to the pathogen causing agglutination where phagocytes can then digest a large number of cells. Once the phagocytes have killed off the pathogens the antibodies decrease due to a short lifespan shown in the dip after primary response.

In the secondary response ^{memory} B plasma cells can differentiate quickly ^{into} ^{b plasma cells which produce} antibodies allowing for a quicker response with a higher overall concentration which also makes the overall response shorter compared to the primary.

Question 5

Many responses to this question were at Level 1. The question was focused on the specification point that learners should know that cell theory is a unifying concept stating that cells are a fundamental unit of structure, function and organisation in all living organisms. Learners were provided with the instruction below the question directing them to use cell theory and figure 7 to support their answer which should have explained that skin is an

organ and blood is a tissue. Learners were expected to define the levels of organisation, such as different tissue types group together to form organs and similar cell types group together to form tissues that have a specific function. The learners could then use the diagrams of skin and blood in a blood vessel to explain that skin fulfils the definition of an organ and blood fulfils the definition of a tissue.

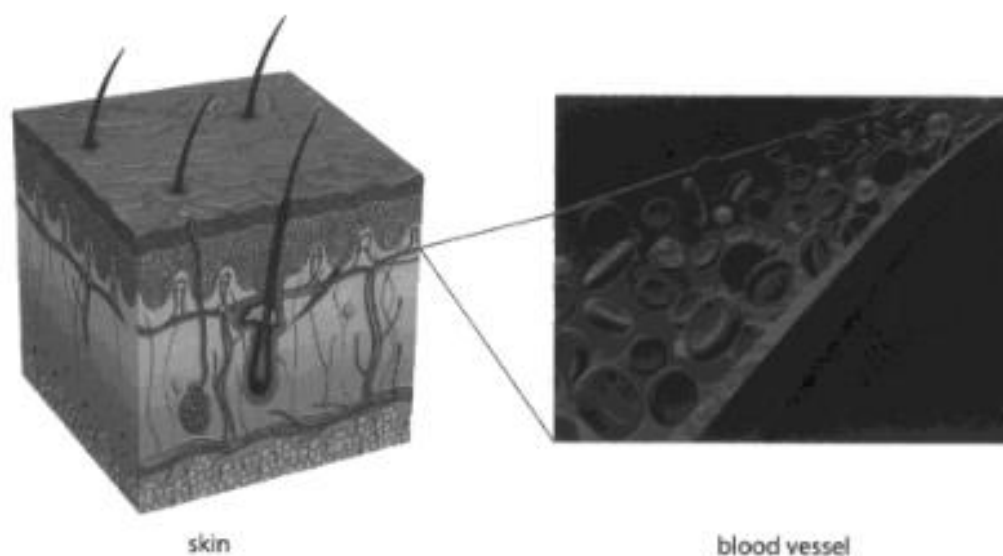
Unfortunately cell theory was frequently overlooked, and many learners misunderstood the requirements of the question and attempted to describe skin and blood vessels, often demonstrating knowledge beyond that expected from the specification. The references to the structure and functions of skin and blood could be credited as a partial response to the level-based question. Responses often included descriptions of blood cell adaptations or their functions, general eukaryotic cell structure, or referenced unrelated examples such as root hair cells and cell walls. Few responses recognised blood as a tissue or skin as an organ.

The following response was a level 1 response and was awarded 1 mark. The learner has given the start to an explanation in that they have only referenced blood; there is no mention of skin. They have given some information about the function of the cells within blood and therefore this is creditworthy.

5 Multicellular organisms have different levels of organisation.

The levels of organisation can be defined by cell theory.

Figure 7 shows cross sections through the skin and a blood vessel in the skin.



(Source: © PAL)

Figure 7

Explain the levels of organisation for skin **and** for blood in the blood vessel.

(6)

Use cell theory and Figure 7 to support your answer.

Blood cells have a biconcave shape which allows for a large surface area and therefore the oxygen that binds to chemical haemoglobin can be taken to the blood stream. And their shape allows them to ~~seper~~ squeeze through capillaries as they are flexible and this allow for more blood flow and oxygen to go around the body.

The following response was a level 2 response and was awarded 3 marks. This was a common approach to this question. The learners have focused on the figure rather than the question and have given a lot of information to describe the function of blood. As this response covers several points in the indicative content and could be used to support the conclusion that blood is a tissue because it is made of different types of cells this response was

creditworthy, even though at first glance it does not seem to be the right topic for the question. The learner has also made reference to skin is an organ and has given a definition. Therefore overall this is sufficient to award a mark in level 2.

Explain the levels of organisation for skin **and** for blood in the blood vessel.

(6)

Use cell theory and Figure 7 to support your answer.

blood

* has white and red blood cells.

↳ red- biconcave- allows oxygen to flow through the blood

↳ white- ^{contain} neutrophils which allows them to squeeze through gaps to reach the infection site

- lymphocytes- contain T and B cells.

- T cells send signals to B cells and B cells produce pathogens. - T cells ingest

cancerous cells.

↳ arteries carry blood away from the heart

↳ veins carry blood to the heart.

↳ capillaries

skin

↳ Organ- group of tissues joined together for a specific function

↳ contains hair cells

↳ veins

↳ arteries.

↳

The following response was a level 3 response and was awarded 5 marks. The response has focused on the definitions within cell theory. The learner has correctly stated that skin is an organ and provided a definition of an organ. The learner has stated that blood is made up of different cells but has not used this to explain that therefore blood is a tissue. This would have given the learner six marks.

Explain the levels of organisation for skin **and** for blood in the blood vessel.

(6)

Use cell theory and Figure 7 to support your answer.

- Cell theory is the concept that cells build up to make tissues, tissues build up to make organs, organs ~~build~~ make organ systems.
- ~~Blood~~ Skin is an ~~organism~~ organ built up of tissues.
- Red blood cells are in the blood vessel.
- Red blood cells have large surface area.
- Skin is built of different layers (figure 7).
- Blood vessel has different types of blood cells (white, red, etc).
- White blood cell ^{engulfs} ingests pathogens.
- ^{the} Skin has blood vessels inside which contain blood cells.
- Cells make up the tissues to ~~from~~ form blood vessels and tissues build to make the organ (skin).
- Skin is largest organ in the body.

Chemistry

Question 1

The focus of this question was carbon and its compounds.

1(a)

This question asked learners to state the group and period of the periodic table where carbon is found. A good proportion of learners scored on the question for knowing that carbon is group 4 and period 2. Where learners did not score, it was often because they gave the period as 1 or 14 rather than 2. In some cases, learners gave 6 rather than 4 or wrote non-metal as the group rather than the group number, showing a misunderstanding of the question.

This shows a good answer which gained the mark.

1 Carbon is an element in the periodic table.

(a) State the group and period of the periodic table where carbon is found.

(1)

group = 4
period = 2

Some learners gave 14 as the group, this was accepted.

1 Carbon is an element in the periodic table.

(a) State the group and period of the periodic table where carbon is found.

(1)

group = 14
period = 2

A common incorrect answer seen was where learners gave 4 as the group number and 14 as the period. As both numbers were required to be correct for the mark, this example scored no marks.

1 Carbon is an element in the periodic table.

(a) State the group and period of the periodic table where carbon is found.

(1)

group = 4
period = 14

1(b)(i)

This question told learners that carbon reacts with oxygen in a combustion reaction. They were asked in part (i) to write the symbol equation for the reaction. This question was well attempted with a good proportion of learners gaining the mark for the correct symbol equation as in this example.

(b) Carbon reacts with oxygen to form carbon dioxide in a combustion reaction.

(i) Write the symbol equation for this reaction.

(1)



Where learners did not score, it was often because they did not recall that oxygen is diatomic. This example scored 0 marks.

(b) Carbon reacts with oxygen to form carbon dioxide in a combustion reaction.

(i) Write the symbol equation for this reaction.

(1)



Some learners knew the chemistry but lost the mark as they did not use the correct scientific conventions for writing formula. This example scored no marks. It was noted that in some typed scripts, learners did not attempt to change the 2 to subscript. It was not clear if they knew that it should be subscript, or they did not know how to access the function on the computer. Centres should ensure that learners that are using a computer to type answers are able to use the superscript and subscript functions for their chemistry examinations.

(b) Carbon reacts with oxygen to form carbon dioxide in a combustion reaction.

(i) Write the symbol equation for this reaction.

(1)



In some cases, learners did not read the question and gave the formula for carbon dioxide, rather than the symbol equation for the reaction, this scored no marks.

(b) Carbon reacts with oxygen to form carbon dioxide in a combustion reaction.

(i) Write the symbol equation for this reaction.

(1)



1(b)(ii)

This asked learners to explain why carbon monoxide can also be formed during combustion reactions. This proved much more challenging with fewer learners scoring and only the most able scoring the two marks available for understanding that carbon monoxide may be formed when incomplete combustion takes place if there is insufficient or a lack of oxygen. Many learners stated that carbon monoxide was toxic which, although it did not gain credit as it did not answer the question, it did show that there was a general familiarity with carbon monoxide, just a lack of understanding of how it is formed.

Where learners did score, they often scored for stating that there was a lack of oxygen, only the most able could name the process as incomplete combustion, as in this example which scored both marks.

(ii) Explain why carbon monoxide can also be formed during combustion reactions of carbon with oxygen.

(2)

This happens when it is incomplete combustion where there isn't enough O_2

Many learners misinterpreted the question and stated that it was formed because oxygen and carbon react or by focusing on the numbers of carbon and oxygen atoms involved in the reaction, often stating that carbon reacts with just one oxygen. A noticeable proportion referred to carbon monoxide as CO_3 .

Some learners described the reaction rather than explaining it, this example gained 1 mark for stating that it is incomplete combustion that causes carbon monoxide. The answer then describes the carbon bonding to the oxygen with a triple bond, instead of explaining that it is due to the lack of oxygen. Centres should ensure that their learners are familiar with the

different command words and used in the specification and what is required by each.

(ii) Explain why carbon monoxide can also be formed during combustion reactions of carbon with oxygen.

(2)

The Incomplete combustion is what causes carbon monoxide to form. This is when one carbon and one oxygen atom are triple bonded.

Q1(c)(i)

This question started by showing learners a molecule of carbon dioxide and a molecule of carbon monoxide. They were told that carbon dioxide contains double bonds and that carbon monoxide contains triple bonds. Learners were asked to explain the difference in strength between the carbon to oxygen bonds in the two molecules.

Learners accessed the question well, with the majority understanding that the triple bond, or the bond in the carbon monoxide molecule, would be stronger. However, they then struggled to explain why that bond was stronger, with only the most able understanding that there were fewer shared pairs of electrons in the double bond or vice versa. A common misconception seen was the carbon monoxide had stronger intermolecular forces. This was rejected against the second marking point.

This example shows a good example which scored both marks for stating that the carbon monoxide is stronger and that this is because it has more shared pairs of electrons.

(c) Figure 1 shows a molecule of carbon dioxide and a molecule of carbon monoxide.



carbon dioxide



carbon monoxide

Figure 1

Carbon dioxide contains double bonds.

Carbon monoxide contains a triple bond.

- (i) Explain the difference in strength between the carbon to oxygen bonds in the two molecules shown in Figure 1.

(2)

The carbon monoxide bond is stronger as it has a triple bond between the carbon and oxygen meaning there is more shared pairs of electrons, strengthening the bond. Carbon dioxide only has 2 double bonds.

This next example scored 1 mark. The learner shows an understanding that the carbon dioxide bonds take less energy to break than the carbon monoxide bonds, this was accepted for the first mark. However, they explain this by saying that the carbon monoxide has stronger intermolecular forces which is rejected so just 1 mark was awarded.

(c) Figure 1 shows a molecule of carbon dioxide and a molecule of carbon monoxide.



carbon dioxide



carbon monoxide

Figure 1

Carbon dioxide contains double bonds.

Carbon monoxide contains a triple bond.

- (i) Explain the difference in strength between the carbon to oxygen bonds in the two molecules shown in Figure 1.

(2)

Carbon monoxide has stronger intermolecular forces in the bonds as it is a triple bond where as carbon dioxide contains double bonds this shows it requires less energy to break the bonds than carbon monoxide.

1(c)(ii)

Learners found this much more challenging with only the more able scoring, and few scoring the highest marks. Those that did score mostly did so for the observation that oxygen is more electronegative than carbon, the more able knew that this results in electrons being attracted or pulled towards the oxygen. Only the most able knew that the presence of the oxygen either side of the carbon in the carbon dioxide meant that the dipoles cancelled out.

A common misconception seen was where learners referred to the electronegativity of the two molecules rather than the atoms involved in them with many adding up the electronegativity values of the atoms and to try to come to a conclusion. A significant proportion of learners repeated the stem of the question; this gained no credit.

The following example shows a good answer, which gained the full four marks.

Table 1 shows the electronegativity values for carbon and for oxygen.

	electronegativity value
carbon	2.5
oxygen	3.5

Table 1

- (ii) Explain why carbon monoxide has permanent dipole-dipole forces of attraction but carbon dioxide does not.

(4)

Carbon has a lower electronegativity than oxygen. This means that at any point, electrons will be more attracted to the oxygen atom in the bond. In carbon dioxide, the pulls created by the oxygen atoms counteract each other as they are pulling from opposite sides. This means the electrons do not create a dipole. In carbon monoxide, there is only a pull from one oxygen atom from one side. This means the electrons are pulled closer to the oxygen atom causing a permanent dipole-dipole force.

This example showed a common answer that scored no marks.

Table 1 shows the electronegativity values for carbon and for oxygen.

	electronegativity value
carbon	2.5
oxygen	3.5

Table 1

(ii) Explain why carbon monoxide has permanent dipole-dipole forces of attraction but carbon dioxide does not.

(4)

Carbon monoxide has permanent dipole-dipole forces because it has a lower electronegativity value meaning it's tendency to gain an electron when reacting is easier. Permanent dipole-dipole forces are polar molecules meaning their electrons are evenly spread out.

Question 2

The focus of this question was on copper.

2(a)(i)

The stem of this question told learners that copper metal is used in electrical wiring and that one reason for that is its ability to conduct an electrical current. The first part of the question was a multiple-choice question, in which learners were asked to identify another reason why copper is used for wiring. In general, learners performed well in the question with many gaining the mark for knowing that copper is ductile.

2(a)(ii)

This asked learners to describe how copper conducts an electrical current. A good proportion scored at least one mark for stating that copper contains delocalised or free electrons, of those that knew that there were delocalised electrons, around half went on to describe the movement of these electrons to gain the second mark.

This example gained both marks available. The electronic configuration was ignored and the marks were awarded for the description of free electrons being able to move.

(ii) Describe how copper conducts an electrical current.

(2)

copper has 29 electrons, 2,8,8,8,3 → The last shell isn't full - the free delocalised electrons are able to move through the structure and carry a charge/electrical current.

A common misconception was that copper contained ions that were free to move, this was rejected for both marking points.

(ii) Describe how copper conducts an electrical current.

(2)

~~The ions in the copper are free to move~~
When molten the ions are free to move and are able to carry the electrical charge.

Q2(b)

Learners were given two natural isotopes of copper and the percentage of each in a sample of copper. They were then asked to calculate the relative atomic mass of this sample of copper. Where learners had practiced calculating relative formula masses from isotopic data they performed well, often showing their working in a clear and logical manner. Different variations of the same method were all awarded credit. However, there were a large proportion that showed a lack of knowledge of the idea of elements having a relative atomic mass based on the average mass of their isotopes. Many took the atomic mass for copper from the periodic table, instead of the values for ^{63}Cu and ^{65}Cu given in the question. Some learners miscalculated the averages by simply multiplying the percentages in the question together.

This shows a common incorrect answer, where learners used the relative atomic mass of copper from the periodic table and multiplied this by the masses of the isotopes given in the question showing a misunderstanding of the calculation required, this gained no marks.

Copper has two natural isotopes, copper-63 and copper-65.

A sample of copper contains 69% copper-63 and 31% copper-65.

(b) Calculate the relative atomic mass of this sample of copper.

(2)

Show your working. $A_r = 63.5$

$$63.5 \times 69 = 4380$$

$$63.5 \times 65 = 4127.5$$

relative atomic mass = 63.5

This example gained 2 marks, it is important to note that 64 alone would score no marks, but as the learner had shown their working, which had the correct method and evaluation, both marks could be scored. The learner has rounded their working correctly, if they had rounded down to 63, the evaluation mark would not have been awarded. This shows the importance of showing your working.

Copper has two natural isotopes, copper-63 and copper-65.

A sample of copper contains 69% copper-63 and 31% copper-65.

(b) Calculate the relative atomic mass of this sample of copper.

(2)

Show your working.

$$(69 \times 63) + (31 \times 65) = 6362$$

$$\frac{6362}{100} = 63.62 = 64$$

~~= 63.62~~

relative atomic mass = 64

2(c)

Learners were given the equation for the production of copper from copper oxide and hydrogen. They were asked to determine the limiting reagent in the reaction when 20g of copper oxide is reacted with 1g of hydrogen. Learners found this more challenging. The most able learners carried out the calculation in a logical way, showing their working at each point to determine that copper oxide was the limiting reagent.

A significant proportion of learners lost a mark as they used 1 as the relative mass of hydrogen rather than 2. Where this happened, error carried forward was accepted and the second and third marking points could be awarded. It was clear that some learners have a misunderstanding of the term limiting reagent, with a noticeable proportion placing a numerical value rather than the name of a reagent on the answer line.

This example scored 2 marks; the learner calculated the relative formula mass of both copper oxide and hydrogen to gain the first mark. Unfortunately, they then have inverted the division to calculate the number of moles and so did not gain the second mark. As the learner has shown their working, error carried forward was allowed and from their incorrect number of moles, they have correctly given hydrogen as the limiting reagent, and the third marking point was awarded.

Copper oxide can be reduced to copper metal using hydrogen.

The equation for the reaction is



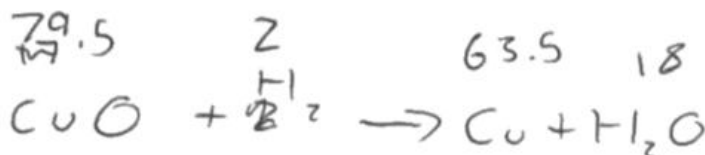
20 g of copper oxide is reacted with 1 g of hydrogen.

(c) Determine the limiting reagent in this reaction.

(relative atomic masses Cu = 63.5, O = 16.0, H = 1.0)

(3)

Show your working.



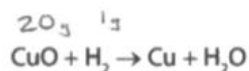
$$\begin{array}{l} 79.5 \\ \div 20 \\ = 3.975 \end{array} \quad \begin{array}{l} 2 \\ \div 1 \\ = 2 \end{array} \quad \swarrow \text{1g more needed}$$

limiting reagent = Hydrogen

This example scored the full three marks; the learner has shown a clear and logical method to come to the correct limiting reagent.

Copper oxide can be reduced to copper metal using hydrogen.

The equation for the reaction is



20 g of copper oxide is reacted with 1 g of hydrogen.

(c) Determine the limiting reagent in this reaction.

(relative atomic masses Cu = 63.5, O = 16.0, H = 1.0)

(3)

Show your working.

$$\text{moles} = \frac{\text{mass}}{\text{mr}}$$

$$\text{moles} = \frac{20}{79.5}$$

$$= 0.25$$

$$\text{mr CuO} = 63.5 + 16 = 79.5$$

$$\text{mr H}_2 = (1 \times 2) = 2$$

$$\text{moles} = \frac{1}{2}$$

$$= 0.5$$

limiting reagent = copper oxide - CuO

Question 3

The focus of this question was scandium and one of its compounds, scandium sulfate.

3(a)

Learners were told that an atom of scandium has 21 electrons and gave part of the electronic configuration of an atom of scandium. They were asked to complete the electronic configuration.

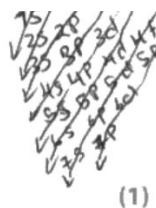
Learners found this quite challenging and whilst some learners correctly identified the electron configuration, there were a wide variety of incorrect answers seen, some of the most common being $4s^2 4d^1$; $3d^3$; $4s^2 4p^1$, showing that learners knew how many electrons were needed but not how they were arranged.

The following example scored the mark.

3 Scandium is a metal in the d-block of the periodic table.

An atom of scandium contains 21 electrons.

(a) Complete the electronic configuration for the atom of scandium.



3(b)(i) and (ii)

This looked at the compound scandium sulfate. Part (i) and (ii) were multiple-choice questions.

Learners found it challenging to identify the correct equation for the third ionisation energy of scandium. However, they found identifying the formula of scandium sulfate less so, with more being able to select the correct formula as $Sc_2(SO_4)_3$.

3(b)(iii)

The question asked learners to describe the structure of solid scandium sulfate. Where learners performed well in this question, they gave a clear and concise description of the giant ionic lattice, with strong electrostatic attractions between oppositely charged ions, as in this example which gained 3 marks.

(iii) Describe the structure of solid scandium sulfate.

ionic compound - ^{electrostatic} strong ~~electrostatic~~ forces of attraction between oppositely charged ions.
Giant ionic lattice - 3D structure of metal ions and non-metal ions joined together.

Where learners scored 2 rather than 3 marks, learners were able to describe the ionic lattice but failed to add the detail of the attractions between oppositely charged ions for the final mark as in this example which scored 2 marks.

(iii) Describe the structure of solid scandium sulfate.

(3)

Scandium sulphate has many ionic bonds this means that it will have a giant ionic lattice.

In some cases, learners were not concise with their scientific terms and so lost marks. In this example the learner understands that the structure is ionic and so gained 1 mark. They state that positive and negative atoms, rather than ions are held together. This reference to atoms meant that the second mark point could not be awarded.

(iii) Describe the structure of solid scandium sulfate.

(3)

Scandium sulfate is an ionic bond. it is a giant ionic structure where ~~oppositely~~ positive and negative atoms are held together by electrostatic forces of attraction. The forces of attraction are hard to break so require lots of energy; resulting in high melting and boiling points.

Less able learners often gave confused answers, mixing types of bonding, often making references to structure of covalent, metallic or intermolecular bonding in the same answer thus contradicting relevant ionic statements. This example scored no marks.

(iii) Describe the structure of solid scandium sulfate.

(3)

~~Strong intermolecular~~
giant ionic lattice, which has strong
intermolecular forces, ~~as~~ with strong bonds
between the elements

Question 4

This was the six-mark levels-based question.

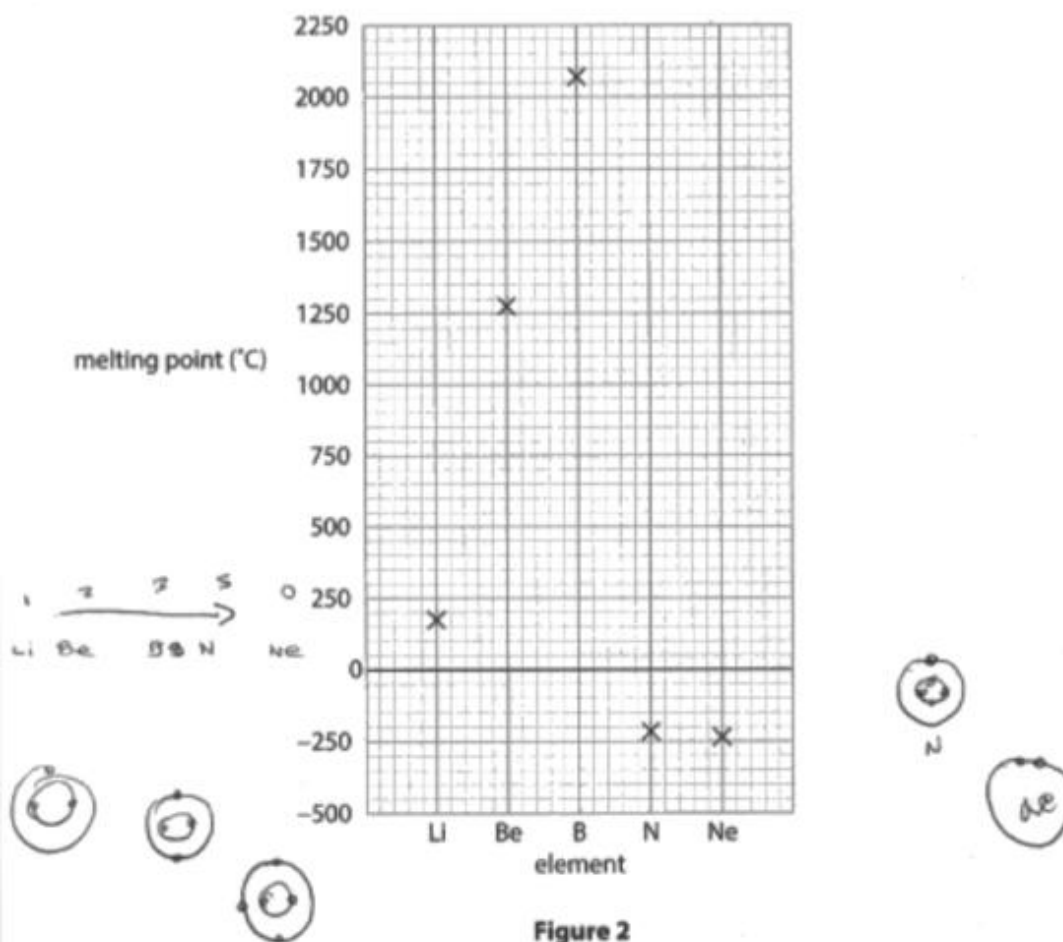
Learners were given a graph showing the melting points of five elements in period two. They were asked to discuss the difference in melting points shown and asked to refer to the type of bonding and structure of the elements in their answer. A good proportion of learners accessed the question which differentiated well across the marks.

The question showed many misconceptions held by learners. A large proportion of learners believed that boron was a metal due to its high melting point. Others thought that all of the elements were ionically bonded or tried to explain all the elements melting points in terms of intermolecular forces or loss of electrons.

Where learners did not score, it was often as they did not read the question carefully and answered a different question on ionisation energies across the period or electronegativity, perhaps after practicing past papers and bringing their knowledge from there. The weaker learners scored credit for understanding how the elements type and bonding changed across the period, but they could not discuss this in any further detail to score marks in level 2 and above.

The following example shows a typical answer where learners had the misconception that the trend in melting point is driving by the ability to lose or gain electrons to obtain a full outer. This gained no marks.

Figure 2 shows a graph of the melting points of some of the elements in Period 2.



4 Discuss the difference in melting points of the elements shown in Figure 2.

In your answer, refer to the type of bonding and structure of the elements.

(6)

Firstly, in Figure 2, the trend is that, as you go across period 2, the melting points of elements Li, Be and B increase. However, when it comes to elements N and Ne, we can see that there is a drastic drop in melting points. This is because Li (Lithium) is in group 1 so has 1 electron on outer shell that needs to be lost in order to have a full outer shell of electrons.

Be (Beryllium) is in group 2 so has to lose 2 electrons to ~~lose~~ gain a full outer shell of electrons and B (boron) is in group 3 so has to ~~lose~~ ^{gain} 3 electrons in order to gain a full outer shell of electrons. The melting points of these elements increase as you go across the period is because each time, ~~there is an additional electron that needs to lose~~ more energy is required to lose/gain electrons. Additionally, N is in group 5 ₀ ~~so it has to lose 5 electrons to get a full outer shell~~ and Ne ~~has to lose 8 electrons~~ so has a full outer shell of electrons. In group 4 (transition metals) the melting points increase as you down the group. ~~because the outer electrons are further away from nucleus so less of~~ ~~etc~~

(Total for Question 4 = 6 marks)

This next example scored 2 marks. The learner correctly identified all elements in terms of metals or non-metals and has given the correct type of bonding for some of the elements. Unfortunately, there is no discussion as to why the melting points of any of the elements are low or high in terms of structure or bonding, therefore the answer remained in level 1 and 2 marks were awarded.

Figure 2 shows a graph of the melting points of some of the elements in Period 2.

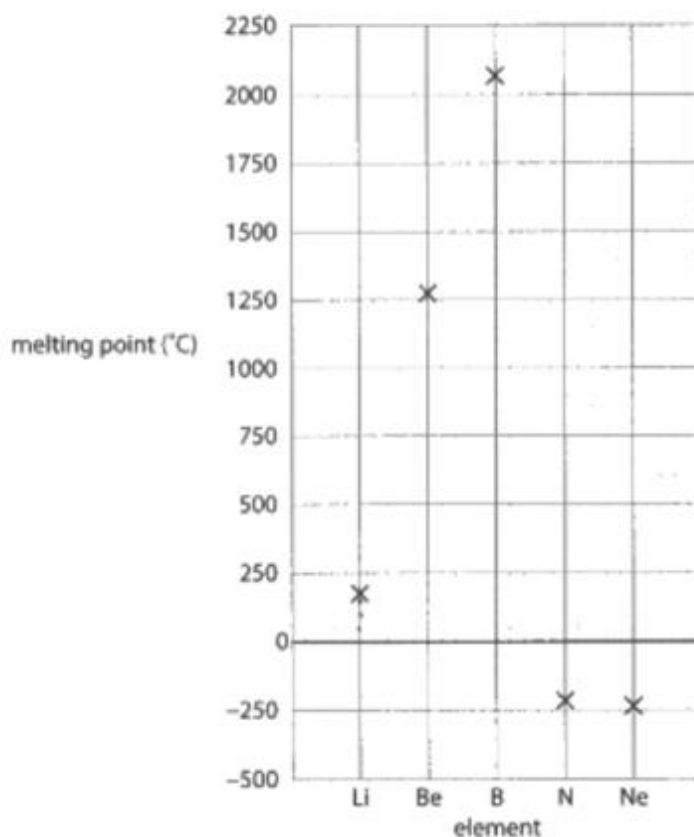


Figure 2

4 Discuss the difference in melting points of the elements shown in Figure 2.

In your answer, refer to the type of bonding and structure of the elements.

(6)

Lithium

- This is an alkali metal
- It has a low melting point at about 175°C
- This is because weak structure

Beryllium

- This is a metal
- It has a high melting point at about 1275°C

• This is because it has a stronger structure than lithium

• It is further away from the nucleus.

Carbon

• This is a non metal

• It has a high melting point of about 2075°C

• This is because it is the hardest of all the elements given in the graph

• It also has covalent bonds

• It is strong because it is ~~connected~~ ^{has} giant covalent bonds

Nitrogen and Neon

• They both have a very low melting point

• It

Nitrogen and Neon

• They are both non-metals

• They ~~are~~ both have simple covalent bonds.

(Total for Question 4 = 6 marks)

The following example shows an answer that scored 4 marks. The learner has initially misread the question and discusses the graph in terms of electronegativity – this was ignored.

On the second page the learner discusses lithium losing 1 electron to the sea of electrons and beryllium losing 2 electrons to the sea of delocalised electrons, they explain that this means that the electrostatic attraction will be higher and more energy would be needed to break the bonds and therefore a higher melting point. Unfortunately, they go on to give the same reasoning for boron. The learner then states that nitrogen and neon have a low melting points but tried to explain this in terms of removal of electrons so this gained no credit. If the learner had discussed the low boiling points of nitrogen or neon or the very high melting point of boron correctly, this could have taken the answer into level 3.

the melting point for Lithium is low, this is because its outermost electron is ~~very~~ easily lost, so it doesn't take as much energy to break the bonds, even though its ^{outer electron} attraction to the nucleus is high. This also means that Lithium only loses 1 electron into the sea of electrons and so with Beryllium ~~losing 2~~ having 2 electrons on its outer shell, it will lose more electrons to the sea of electrons and so its electrostatic attraction will be higher and need more energy to break the bonds. Lithium also has a less positive nucleus than Beryllium so Beryllium ^{outer electron} has more electrostatic attraction to its nucleus so it requires more energy to break its bonds so more heat is required so higher melting point. Boron has the highest melting point because it loses 3 electrons to its sea of electrons so has an even higher electrostatic attraction between the ^{positive} nucleus and negative. Boron has 3 protons too, more than Lithium Li and Be so it has ^{stronger} ~~more~~ electrostatic attraction between the positive nuclei and the negative outermost electrons so it needs more energy to break those bonds so the melting point is higher. N has a very low melting point because the energy required to take its outermost electron ~~is~~ is low since the electrostatic attraction is low, this is because it has more shells and increased shielding and radius, so the outermost electrons are very far from the nucleus, making the attraction weak and easy to break. Ne has a very large radius and lots of shielding due to lots more electrons, this causes it to have a low melting point and as there is less energy required to ~~take~~ take the outermost electron so it has the lowest melting point.

(Total for Question 4 = 6 marks)

This last example shows a good example that scored the full 6 marks.

Figure 2 shows a graph of the melting points of some of the elements in Period 2.

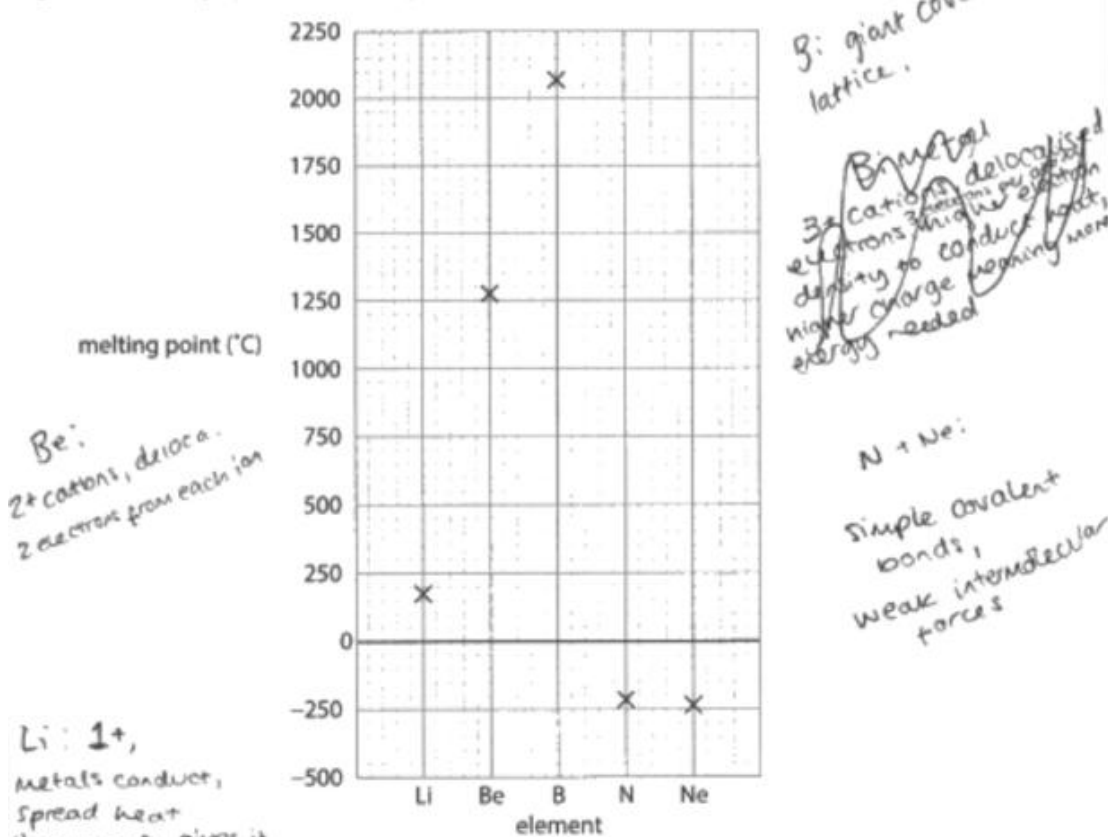


Figure 2

4 Discuss the difference in melting points of the elements shown in Figure 2.

In your answer, refer to the type of bonding and structure of the elements.

(6)

Li is a metal, ~~hence~~ meaning it has metallic bonds. Since Li is in group 1, the cations in the metallic bond have a 1+ charge, and each ion in the lattice will have given up 1 electron. These electrons are delocalised and can move freely with the metal, meaning heat can be carried throughout the entire metal. This thermal conductivity distributes the heat rather than it be concentrated in one area, which allows Li to have a high melting point. Be is also

a metallic bond, however it is in group 2 meaning the atoms form $2+$ cations, and the ions give 2 electrons each to the metallic bond. This creates a higher electron density than a group 1 metallic bond due to the increase in delocalised electrons, and the more electrons to carry heat. The $2+$ charged ions will also have greater electrostatic attractions to the delocalised electrons than group 1 metallic bonds, meaning more energy will be needed to break the forces, explain the increase in melting point from Li to Be. ~~Be forms a giant metallic lattice as it is in group 2.~~ B forms giant covalent lattices as it is in group 3. Covalent bonds are very strong in giant covalent lattices, as there is ~~an~~ electrostatic attraction from all atoms in the lattice, this means a lot of energy is needed to break the covalent bond and melt the structure. Ne and N ~~are~~ both form simple covalent bonds, which have weak intermolecular forces, and the electrostatic attraction is small. Ne and N has small atomic numbers, meaning their nuclear charge is not very high, allowing small amounts of energy to break the forces and melt the elements.

(Total for Question 4 = 6 marks)

Physics

Question 1

1(a)

Most learners were able to recall that water waves are transverse waves.

1 Figure 1 shows a surface wave in a tank of water.

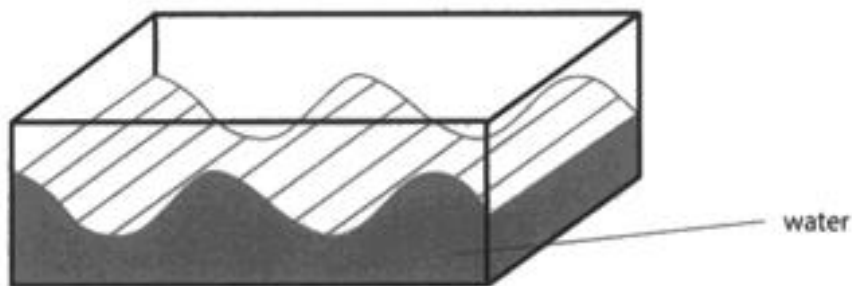


Figure 1

(a) State the type of wave shown on the surface of the water in Figure 1.

(1)

Transverse wave

1(b)(i)

Learners were provided with the diagram shown below and asked to identify the arrow that shows the amplitude of the wave.

(b) Figure 2 shows a graphical representation of the surface wave in the tank.

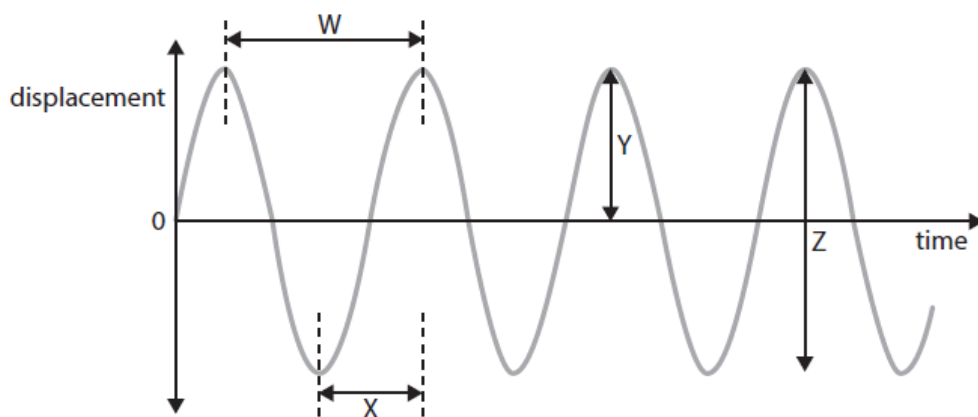


Figure 2

(i) Identify the arrow that shows the amplitude of the wave.

(1)

- ☐ A arrow W
- ☐ B arrow X
- ☐ C arrow Y
- ☐ D arrow Z

The majority of learners correctly chose option C, arrow Y.

1(b)(ii)

From the same diagram, learners were asked to identify the arrow that shows the time period of the wave.

(ii) Identify the arrow that shows the time period of the wave.

(1)

- ☐ A arrow W
- ☐ B arrow X
- ☐ C arrow Y
- ☐ D arrow Z

This proved more challenging, with just over half of learners able to correctly identify option A, arrow W.

Question 2

2(a)

Most learners were able to name the type of signal that consists of an encoded series of zeroes and ones, with either digital or binary acceptable as correct responses. A few learners incorrectly suggested analogue signals.

2 A TV station communicates with a satellite orbiting the Earth by sending a signal.

The signal consists of an encoded series of zeroes (0) and ones (1).

(a) Name the type of signal that consists of an encoded series of zeroes and ones.

(1)

Digital

2 A TV station communicates with a satellite orbiting the Earth by sending a signal.

The signal consists of an encoded series of zeroes (0) and ones (1).

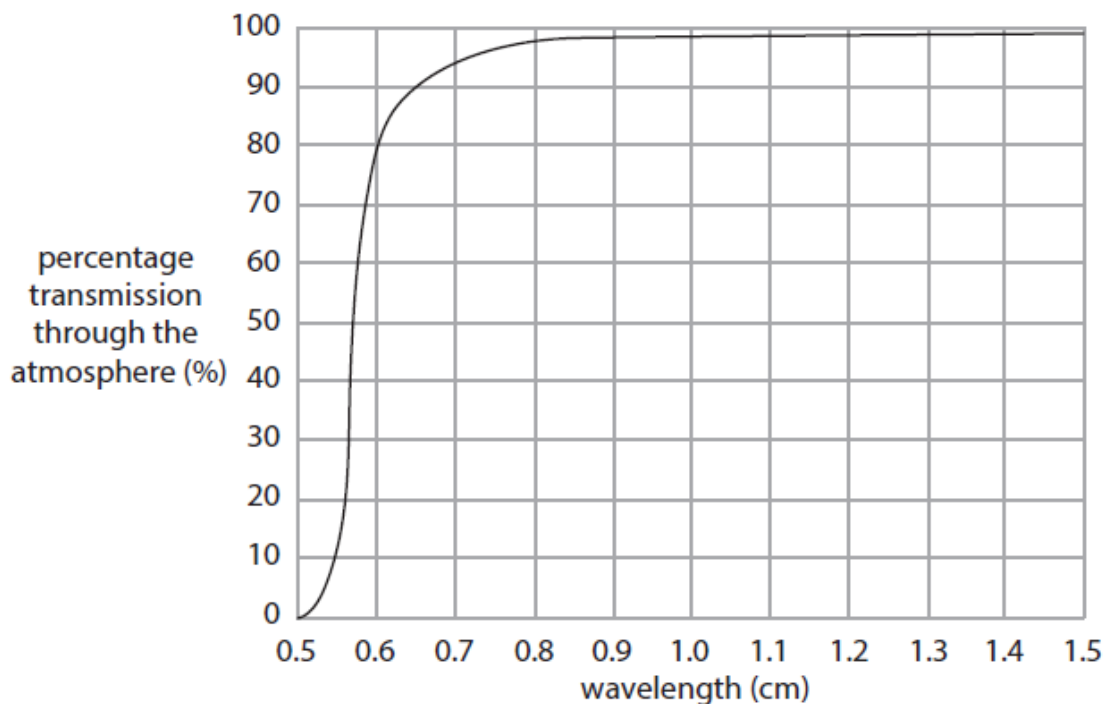
(a) Name the type of signal that consists of an encoded series of zeroes and ones.

(1)

Binary

2(b)

For this question learners were asked to describe the trend shown in the graph below.



The majority of learners were able to achieve at least 1 of the 2 marks available, usually for correctly describing that percentage transmission through the atmosphere increases with increasing wavelength. The example below scored full marks.

As a wavelength in cm increases from 0.5cm to 0.6cm ~~the~~ so does the percentage transmission, from 0 to 80%. Then from 0.6 to 0.7cm it increases slightly from 80% to around 98%. It then stays at 98% as the wavelength increases from 0.8cm to 1.5cm.

Centres should encourage learners to include numerical data as reference points when answering questions of this nature as this ensures an accurate description of the trend shown in the graph.

2(c)

Many learners struggled to explain why low frequency radio waves do not travel out of the atmosphere. Common incorrect responses tended to include vague statements about radio waves not having enough energy, or unable to travel long distances. The response below scored zero marks.

(c) Explain why low frequency radio waves do **not** travel out of the atmosphere.

(2)

it would be because they have a shorter bandwidth with less energy meaning they can't travel long distances.

Some learners did have an understanding that the waves were being reflected, but were not specific in terms of which part of the atmosphere was involved. This response did not score any marks as it did not identify the ionosphere nor explain why the ionosphere is able to reflect low frequency radio waves.

(c) Explain why low frequency radio waves do **not** travel out of the atmosphere.

(2)

Low frequency radio waves don't travel out of the atmosphere because ~~of~~ they are reflected by the atmosphere and they bounce back.

The response below correctly identifies the ionosphere and explains how the ionosphere is able to reflect low frequency radio waves. This response gained full marks.

(c) Explain why low frequency radio waves do **not** travel out of the atmosphere.

(2)

They do not travel out of the atmosphere because of the earth's atmosphere interaction with the ionosphere low frequency waves such as radio waves with very long wavelength. The ionosphere is a layer of the atmosphere with fully charged particles (ions) which reflects low frequency waves back to earth the waves do not have enough energy to pass or penetrate this layer

(Total for Question 2 = 5 marks)

Question 3

3(a)

Most learners were able to describe a ray of light refracting as it passes from air into glass. Many learners were able to expand on their answer and score full marks by describing what happens to the light during this refraction.

3 Figure 4 shows a ray of light travelling from the air into a solid glass cylinder.

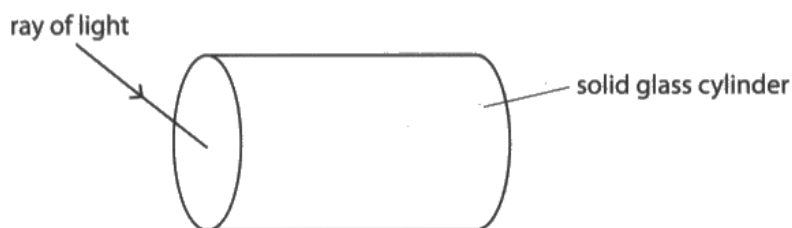


Figure 4

(a) Describe what happens to the ray of light as it travels from the air into the glass cylinder.

(2)

AS light enters a more dense material
it slows down, and will begin
to bend towards the normal line

This response scored both marks; whilst many learners used the correct term 'refract' to describe the behaviour of the ray of light, 'bend' or 'change direction' is also acceptable. This response describes both what happens to the speed of the ray and that it bends towards the normal- either of these is acceptable for the second marking point to be awarded.

Some learners incorrectly thought that the ray of light would diffract, or undergo total internal reflection.

(a) Describe what happens to the ray of light as it travels from the air into the glass cylinder.

(2)

ray of light will diffract as it travels from air to the glass cylinder.

It is recommended that centres ensure learners understand the meaning of similar sounding terminology and take the time to read the question fully.

3(b)

The majority of learners were able to correctly calculate the speed of light in glass using the equation provided. The most common mistake was to incorrectly rearrange the equation. The following response gained full marks.

The glass in the cylinder has a refractive index (n) of 1.6.

(b) Calculate the speed of light in the glass (v).

(3)

The speed of light in air (c) is $3.0 \times 10^8 \text{ m s}^{-1}$.

Use the equation

$$n = \frac{c}{v}$$

Show your working.

$$1.6 = \frac{3 \times 10^8}{v} \quad v = \frac{3 \times 10^8}{1.6}$$

speed of light in the glass = 187500000 ms⁻¹

Centres should encourage learners to show their workings when completing calculations, as many can gain marks for the process used, even if their final answer is incorrect. The response below is an example of this.

The glass in the cylinder has a refractive index (n) of 1.6.

(b) Calculate the speed of light in the glass (v).

(3)

The speed of light in air (c) is $3.0 \times 10^8 \text{ ms}^{-1}$.

Use the equation

$$n = \frac{c}{v}$$

Show your working.

$$\begin{aligned} n &= \frac{c}{v} \\ 1.6 &= \frac{3.0 \times 10^8 \text{ ms}^{-1}}{v} \\ v &= \frac{1.6}{3.0 \times 10^8 \text{ ms}^{-1}} = 5.3 \times 10^{-9} \end{aligned}$$

speed of light in the glass = $5.3 \times 10^{-9} \text{ ms}^{-1}$

Although the value on the answer line is incorrect, this response gained 2 marks as the learner had clearly shown their workings and could be awarded marks for substitution and evaluation. If the workings had not been present, this response would have gained 0 marks.

3(c)

Whilst some learners were able to correctly state that endoscopes use the process of total internal reflection, only a minority were able to fully explain how an image of inside the body is produced. Many responses suggested the use of a single optical fibre, rather than bundles, despite this being mentioned in the question stem. Some learners suggested endoscopes used waves from other regions of the electromagnetic spectrum, for example, X-rays or gamma rays, rather than visible light rays. The response below includes an excellent response to the question and was awarded full marks.

(c) Explain how an image of the stomach lining is produced in the eyepiece of the endoscope.

(4)

Optical fibre is a very fine made of glass. Each fibre sends a pixel and converts the image to be seen on the screen. There are two bundles in the optical fibre, coherent and incoherent. Incoherent bundle have the light that goes inside the stomach for observation, the stomach lining reflects the image back through the coherent bundle which allows the image to be seen on the screen. Total internal reflection happens because the angle of incidence is greater than the critical angle.

Many responses were very vague and did not demonstrate a clear understanding of the process involved. The response below was fairly typical of many which failed to score any marks.

Solid glass cylinders can be used as optical fibres.

A medical endoscope has a thin tube containing two bundles of optical fibres.

The thin tube is inserted through a patient's mouth and into the stomach.

This enables an image of the stomach lining to be produced in the eyepiece of the endoscope.

(c) Explain how an image of the stomach lining is produced in the eyepiece of the endoscope.

(4)

The endoscope sends waves through the stomach and these waves bounce off the stomach. The way the waves are bouncing back determines the image and how it is produced. Once the waves reflect back to the endoscope, the endoscope then produces an accurate image which is shown.

Question 4

4(a)

Most learners were able to suggest that the sensor closest to the signal transmitter would have a higher intensity reading than that of a sensor further away, but few were able to explain why this would happen.

The response below scored both marks.

- (a) Explain why the intensity reading on sensor A is higher than the intensity reading on sensor B.

(2)

intensity decreases as distance increases, a wave is emitted radiated out ward from a point source, its energy spreads over an increasingly larger area intensity is inversely proportional to the square of the distance. A is closer than B, so intensity is higher

$$I = \frac{k}{r^2}$$

Many learners simply repeated part of the question stem rather than trying to give an explanation of why sensor A got a stronger signal and so scored 1 out of the 2 available marks. An example of this type of response is shown below.

- (a) Explain why the intensity reading on sensor A is higher than the intensity reading on sensor B.

(2)

bluetooth signals only travel 10metres so the intensity of the wave would decrease as the distance increased so sensor A is gets a stronger signal whilst sensor B gets a weaker signal

Centres are encouraged to check learners' understanding of the meaning of command words such as explain, and to ensure learners realise a reason or reasons why have to be given in the response when the command word of a question is 'explain'.

Another common misconception seen was to say that the waves are spreading out, perhaps in response to the image in the question. It should be noted, that it is the energy or signal which is spreading out, rather than the wave per se.

4(b)

Many learners were able to use the equation provided in the question to perform some steps of the calculation correctly, gaining at least 2 marks. An example gaining full marks is shown below:

$$\frac{I_A}{I_B} = \frac{(D_B)^2}{(D_A)^2}$$

I is the intensity of the signal at sensor A or B.

D is the distance between the transmitter and sensor A or B.

Show your working.

$$\frac{5.0}{I_B} = \frac{(1.2)^2}{(0.3)^2}$$

$$\frac{5.0}{I_B} = 16$$

~~$$\frac{5.0}{16} = 0.3125$$~~

$$I_B = \frac{5.0}{16} = 0.3125$$

test it

$$\frac{5.0}{0.3125} = 16 \quad \checkmark$$

intensity at Sensor B (I_B) = 0.3125 units

It was particularly pleasing to note this learner had been encouraged to double check the value obtained – this is excellent practice.

Many learners did not correctly calculate the distance of each sensor for the transmitter, but were still able to gain 3 out of the 4 marks available if their workings were shown and the processes of substitution, rearrangement and evaluation had all been correctly performed (error carried forward). The response below is an example of this.

I is the intensity of the signal at sensor A or B.

D is the distance between the transmitter and sensor A or B.

Show your working.

$$\frac{5}{16} = \frac{(1.3)^2}{(0.4)^2} =$$

$$\frac{(1.3)^2}{(0.4)^2} = 10.5625$$

$$5 \div 10.5625 = 0.47337278106508 \dots$$

$$= 0.47$$

intensity at Sensor B (I_B) = 0.47 units

Question 5

5(a)

Most learners were unable to correctly select option D, mass per unit length, in response to this question.

5 Some musical instruments use waves on strings to create sound.

The equation shows the speed of a wave on a string.

$$v = \sqrt{\frac{T}{\mu}}$$

(a) Which quantity is represented by μ in the equation?

- ☐ A amplitude per unit length
- ☐ B force per unit length
- ☐ C frequency per unit length
- ☐ D mass per unit length

Centres are encouraged to make sure learners are familiar with the symbols and equations associated with this specification.

5(b)

This question required learners to apply their knowledge of waves and harmonics to discuss how pulling a bow across a violin's strings makes a sound. Many learners were able to make simple statements about sound waves or violin strings vibrating but were unable to link these with ideas around the creation of stationary waves and the concept of harmonics.

The following response met the level 1 descriptors. The learner has made a series of statements, which are correct e.g. 'strings vibrate', 'sound waves are longitudinal', 'creates a stationary wave'

There has been an attempt to identify the existence of a stationary wave but no correct explanation as to how it is created (an attempt was made but it is weak and missing key points). Despite the fact that this response contains several correct statements, the absence of linkages means this is still a level 1 response with 2 marks awarded.

When pulling the bow across the violin strings, the strings vibrate. This creates a stationary wave and sound can be heard.

The f-shaped amplifies the sound as it allows waves, produced from the string, to travel into the hollow wooden body, which increases the volume of sound.

Sound waves are longitudinal. When pulling the bow across the violin, it creates tension in the strings.

The next response contains a number of links made and the correct use of technical terminology. It was thus placed into level 2 and awarded 4 marks.

The bow is made of a material that can make the strings vibrate. When pulled across the ~~the~~ strings, ~~the~~ the strings vibrate creating a transverse wave which travels down the strings. ~~which~~ ~~each~~ since both ends of the string are closed the ~~note~~ vibrations create a standing wave. The waves bounce back and forth from each ~~the~~ end creating constructive and destructive interference on the wave. The ~~destructive~~ ~~constructive~~ constructive interference create bigger amplitudes which when in contact with the air particles it ~~travels~~ vibrates the air particles creating a ~~transverse~~ ^{long} waves which enter your ear and vibrate your ear drum to the same frequency of the original ~~transverse~~ wave.

This response contains several links e.g. 'since both ends of the string are closed (presume means fixed) the vibrations create a standing wave', 'waves bounce back and forth from each end creating constructive and destructive interference', 'constructive interference create(s) bigger amplitudes'. Many learners made reference to how the human ear hears sound- this is not relevant to the question and so is not creditworthy.

Whilst the section on stationary waves is of good quality, there is little relation of this to how the violin produces a sound, which stops this getting into level 3.

The next response below contains several links e.g. 'longitudinal waves are produced. This is because the oscillation is parallel to the direction the wave is travelling', 'hits a fixed point causing the wave to bounce back (links this to stationary wave formation)', 'constructive is an antinode and destructive is a node', 'antinode produces large amplitude', 'node is when amplitude is zero'. Further links made to waves being in phase for constructive interference and a brief discussion of harmonics.

The first part of the response states that 'when the bow is pulled across the strings, longitudinal waves are produced' – this is incorrect, as it would produce a transverse wave, BUT the sound wave IS longitudinal. The learner was given the benefit of the doubt here as it was unclear what their meaning was. This doesn't detract from the quality of the rest of the answer.

This response is level 3, 6 marks

When the bow is pulled across the strings, longitudinal waves are produced. This is because the oscillation is parallel to the direction the wave is travelling. The wave can produce harmonics and notes which can have a different frequency. As ~~the~~ when the bow is pulled across the strings, ~~this~~ this can produce longitudinal stationary waves. This is by the wave traveling ~~down~~ along and hits a fixed point causing the wave to bounce back. These waves can ~~interfere~~ interfere with waves traveling opposite direction. This can produce constructive or destructive ~~interferences~~ ^{antinodes} antinodes. Constructive is ~~antinode~~ and destructive is a node. ~~Antinodes~~

produces large amplitude and oscillation is high.
~~This means the~~ Node is when amplitude is zero and ~~no~~ oscillation. ~~These~~ A f shaped sound hole ~~is~~ can produce a harsh and bright sound ~~because~~ this produces a node. Also ~~can~~ produce a longer wavelength so the lowest fundamental frequency is shorter. Which produce different harmonics and nodes.
 A bow can produce determine the harmonics that occur. So bow can act like a node and antinode. When constructive interferences ~~occur~~ sound can be heard. This is when the waves are in phase. ~~so the~~ This can produce a frequency so the ~~harmonics~~ harmonics are produced. Also by bowing ~~the~~ across the violin. This external force can match the ~~nodes~~ natural frequency of the violin and can cause energy to build up.

Below is another example of a level 3, 6 marks response. Again, the learner has made a number of links and has provided a comprehensive discussion on how sound is produced from the starting point of the bow being pulled across the violin strings, with a clear description of how a stationary wave is formed, the significance of nodes and antinodes, plus links to the concept of harmonics and how they are related to the pitch of sound produced.

When the bow is pulled along the string, it causes the string to oscillate. This oscillation then travels down the string and is then reflected back along the string. The original wave and the reflected wave superpose, causing them to interfere. Nodes and antinodes are then created along the string, where nodes are points of destructive interference and antinodes are points of constructive interference. At nodes, waves destructively interfere as they are in antiphase, causing the amplitudes of both waves to interfere and cancel each other out. The resulting amplitude at a node is 0, causing no sound to be created, whereas at antinodes, the amplitudes interfere and produce a larger amplitude which emits ^{a vibration} ~~sound~~. This vibration is then echoed within the violin's hollow body, and the sound is emitted through the sound holes. The pitch of the sound ~~created~~ created is also determined by the number of harmonics. If the player places his finger on the string, he will create a new node, causing the string to reflect from that point and create new nodes and antinodes. The more harmonics there are, the higher the pitch.

Summary

Biology

Centres need to fully prepare learners for the exam by practicing exam technique, especially in relation to reading the question carefully and not repeating the stem of the question. Learners should also be taught that when they have answered the question to reread their response to ensure that the question set has been addressed in the answer they have given, and that they have used appropriate scientific knowledge and vocabulary.

Based on their performance on this paper, learners should:

- use the additional guidance to understand the depth and breadth of the specification
- understand the demand of the command verbs, especially the difference between 'describe' and 'explain'
- show their working out for all calculations to ensure mark can be given for the stages within the calculation, including conversions
- revise key definitions with Level 3 detail, including words linked to cell theory, such as tissue and organ
- have a clear understanding of the difference between 'function' and 'structure'
- be familiar with and recognise the ultrastructure and function of organelles in eukaryotic cells (plant and animal cells) and eukaryotic cells (plant-cell specific).

Chemistry

Based on their performance on this paper, learners should:

- ensure that they are aware of the difference in key terms such as atom, electron and ion
- ensure that they are familiar with the formula of common compounds and diatomic elements in the specification and be able to write these formulae using the correct scientific conventions in terms of cases of letters and subscripts

- practice exam techniques, such as using the information in the stem of the question to inform and help them with their answer and checking that this information is not just repeated as the given answer. Perhaps underlining or highlighting key words to remind them as they are forming their responses
- ensure that they are familiar with the different types of bonding in elements and in compounds
- be taught the difference between the different types of command words such as explain, describe and discuss and what is required by each.

Physics

Based on their performance on this paper, learners should:

- learn how different parts of the atmosphere affect the transmission of different waves of the electromagnetic spectrum
- practice rearranging equations and showing their workings in a clear and logical manner
- learn how an endoscope works to enable it to provide clear images of the inside of the body
- be able to describe how stationary waves form and apply this knowledge to musical instruments.

Please refer and link to the Applied Science Specification and/or Sample Assessment Materials (SAMs) on the BTEC Nationals Applied Science webpage, located [here](#).



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